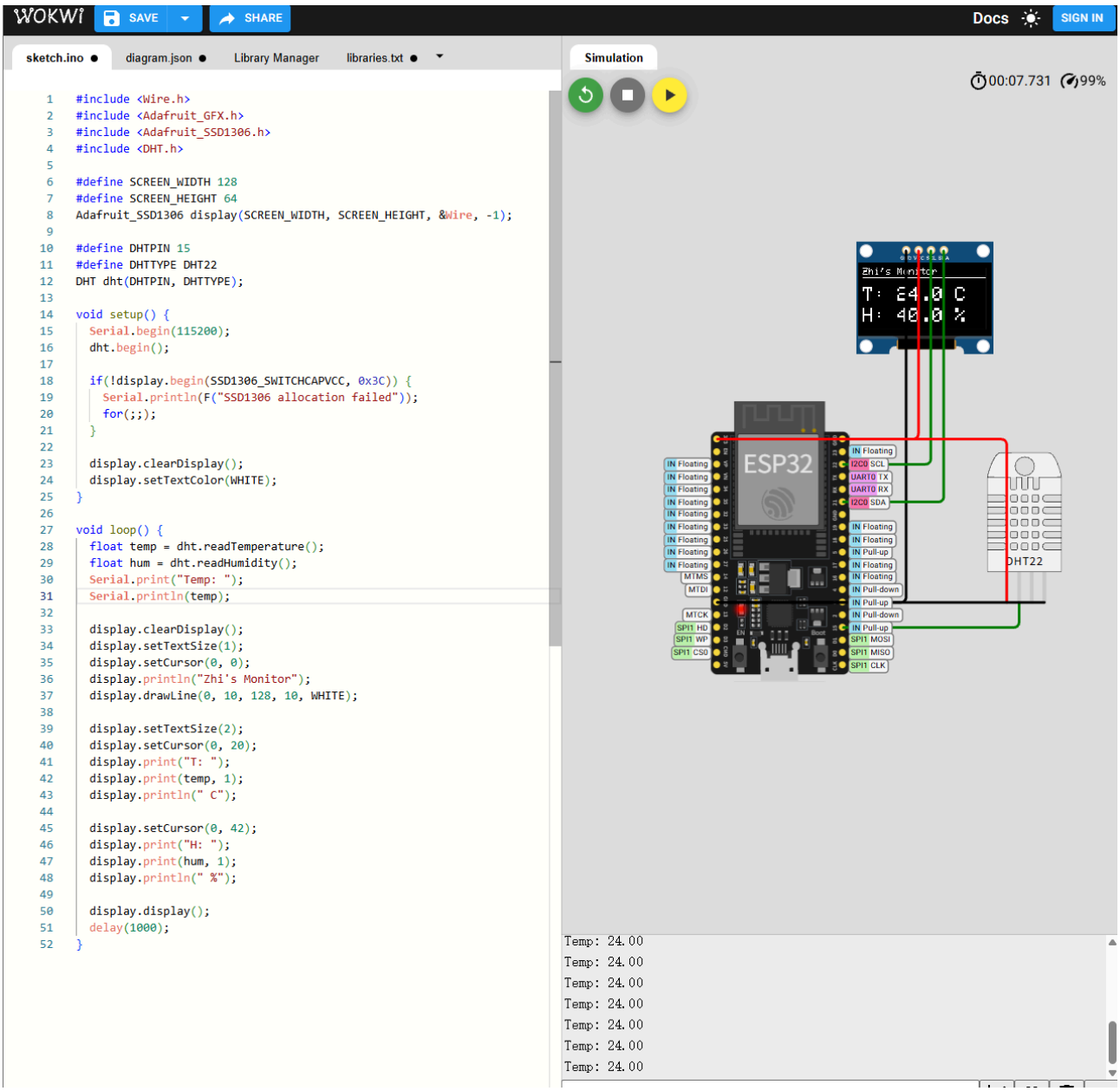
ESP32 IoT Monitor — DHT22 + OLED

A beginner embedded project built in the Wokwi simulator. An ESP32 reads temperature and humidity from a DHT22 sensor and displays the readings in real time on a 128×64 SSD1306 OLED screen, while also printing them to the Serial Monitor.



Overview

- **Microcontroller:** ESP32
- **Display:** SSD1306 0.96" OLED, 128×64, I2C address `0x3C`
- **Sensor:** DHT22 digital temperature & humidity sensor

- **Simulation:** Wokwi (no hardware needed to run)

Features

- Real-time temperature (°C) and humidity (%) readings
- Live OLED display using Adafruit GFX / SSD1306 libraries
- Serial Monitor output at 115200 bps for debugging (`Temp: 24.00`)
- 1-second refresh interval

Wiring / Pinout

Peripheral	Pin	ESP32 Pin	Protocol	Description
SSD1306	VCC	3V3	Power	3.3V supply
SSD1306	GND	GND	Ground	Common ground
SSD1306	SCL	GPIO 22	I2C Clock	I2C clock line
SSD1306	SDA	GPIO 21	I2C Data	I2C data line
DHT22	VCC	3V3	Power	3.3V supply
DHT22	DATA	GPIO 15	1-Wire Digital	Sensor data
DHT22	GND	GND	Ground	Common ground

Project Structure

```
├── sketch.ino       # Main C++ firmware (setup & loop)
├── diagram.json     # Wokwi hardware schematic & wiring
├── libraries.txt    # Library dependencies for Wokwi
├── simulation.png   # Screenshot of running simulation
└── README.md        # Documentation
```

How to Run

Method 1: Web Simulation (no hardware)

1. Import `sketch.ino` and `diagram.json` into Wokwi
2. Add the libraries from `libraries.txt` via Library Manager
3. Click Start Simulation
4. Click the DHT22 module to adjust temperature/humidity sliders and watch the display update

Method 2: Physical Hardware

1. Open the project in Arduino IDE
2. Install: Adafruit SSD1306, Adafruit GFX Library, DHT sensor library
3. Wire the modules using the pinout table above
4. Select ESP32 Dev Module, choose your COM port, and upload

How It Works

- `setup()` runs once at startup: starts Serial (115200), initializes the DHT sensor, and initializes the OLED. If the OLED fails to initialize, the program halts.
- `loop()` repeats forever: reads temperature/humidity, prints temperature to Serial, clears and redraws the OLED, waits 1000 ms.

What I Learned

- Difference between `setup()` (run once, initialization) and `loop()` (runs repeatedly)
- Why a sensor's DATA pin must connect to a GPIO (it sends digital pulses encoding the readings), not to power/ground
- I2C addressing: `0x3C` is the OLED's address on the I2C bus, like a house number that lets the ESP32 talk to the right device

Future Improvements

- [ ] Add LED alarm when temperature exceeds 30°C
- [ ] Show warning message on OLED when threshold is exceeded
- [ ] Add °F display toggle